

The Dynamics of Mercury's Magnetic Boundaries

O. Daniel O'Carroll

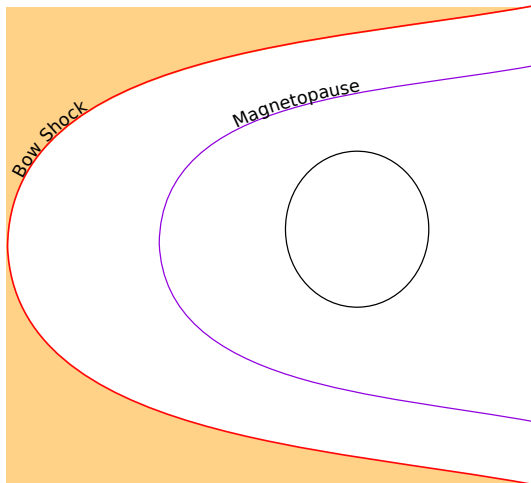
July 3, 2026

Magnetic Field Regions

- Three regions

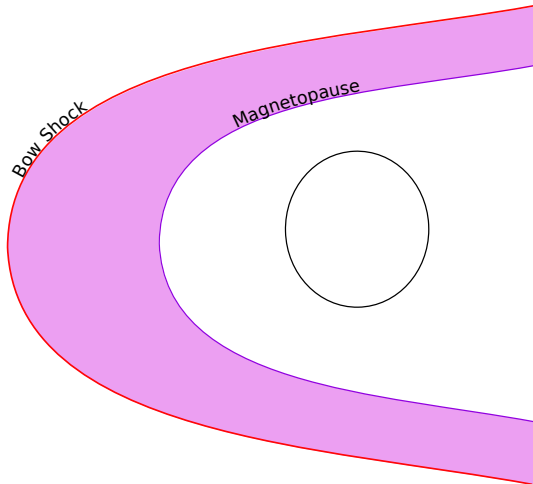
Magnetic Field Regions

- Three regions
 - Solar Wind



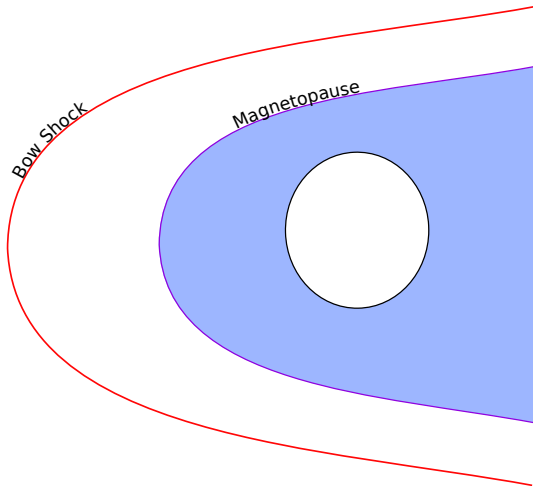
Magnetic Field Regions

- Three regions
 - Solar Wind
 - *Magnetosheath*

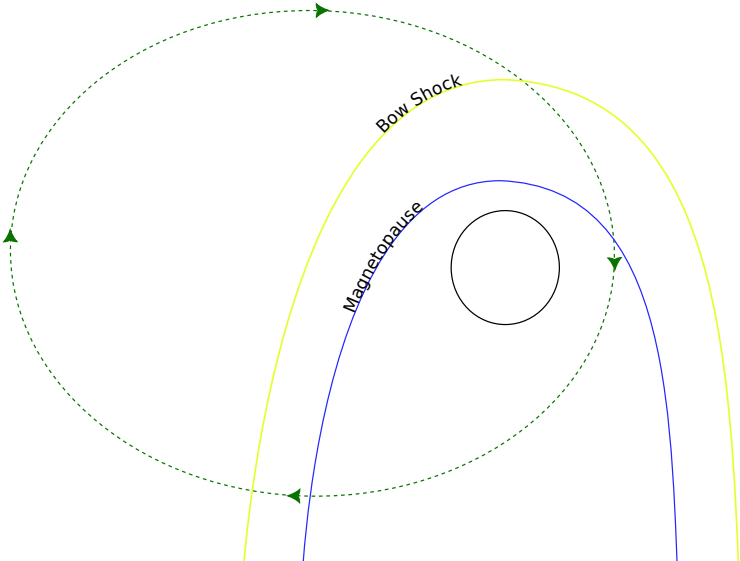


Magnetic Field Regions

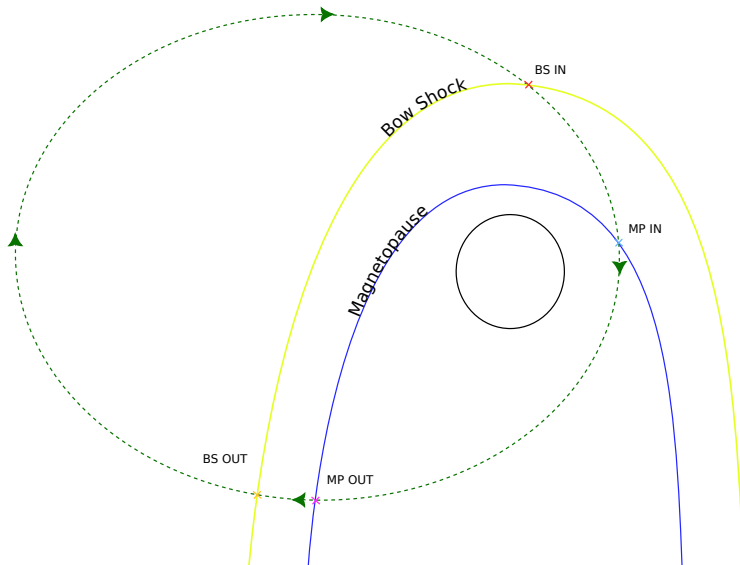
- Three regions
 - Solar Wind
 - *Magnetosheath*
 - Magnetosphere



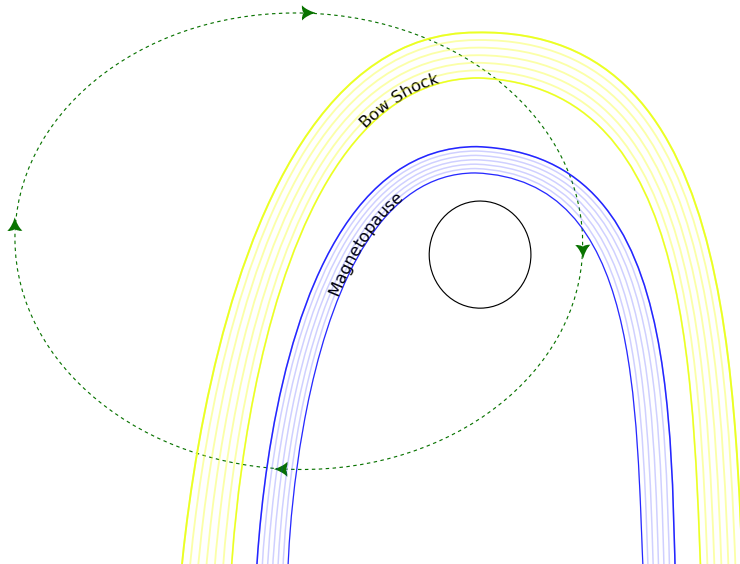
Boundary Crossing



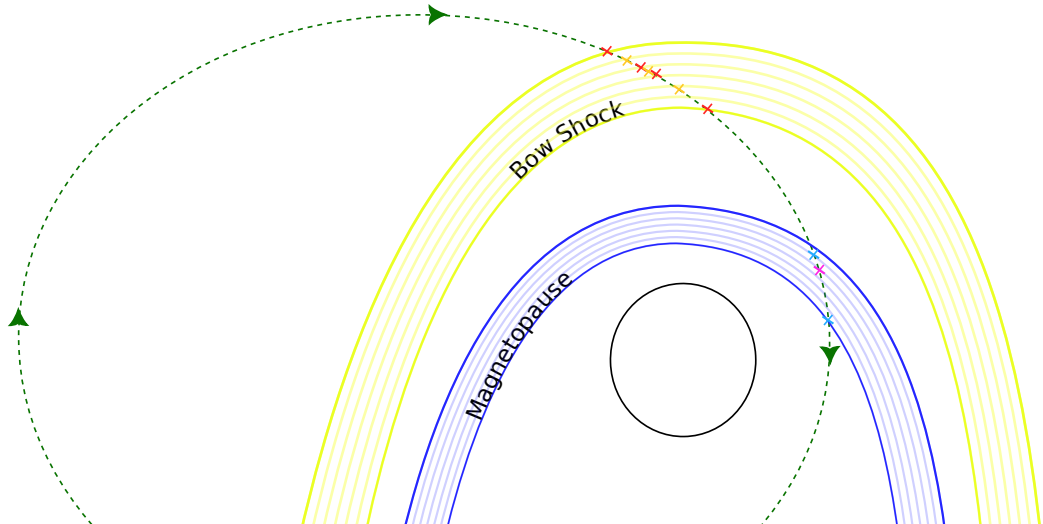
Boundary Crossing



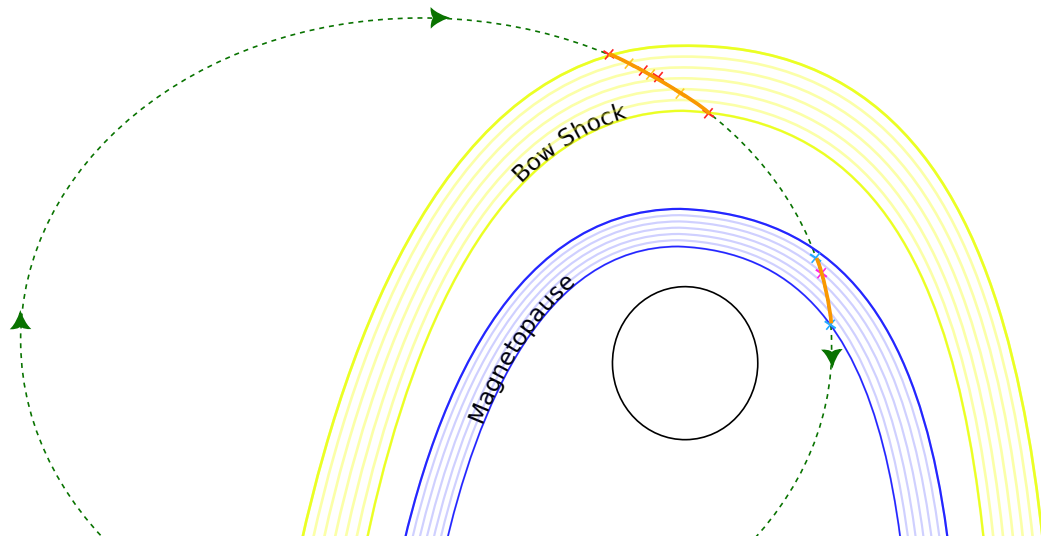
Variability of Boundaries



Variability of Boundaries

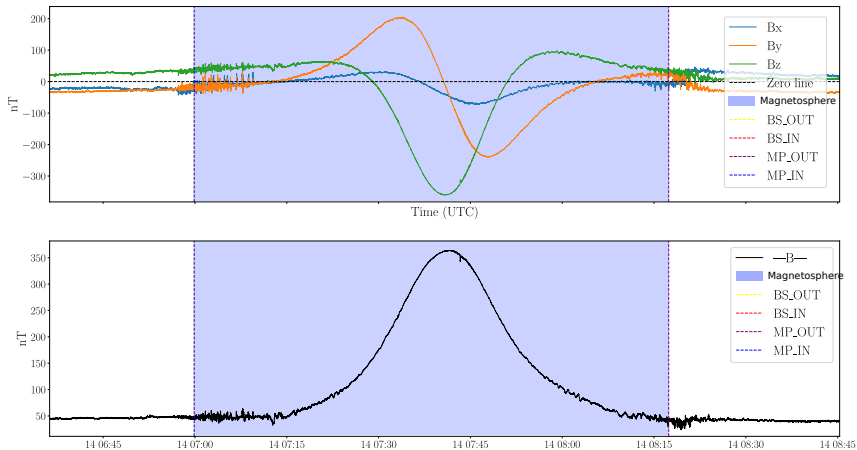


Encounters



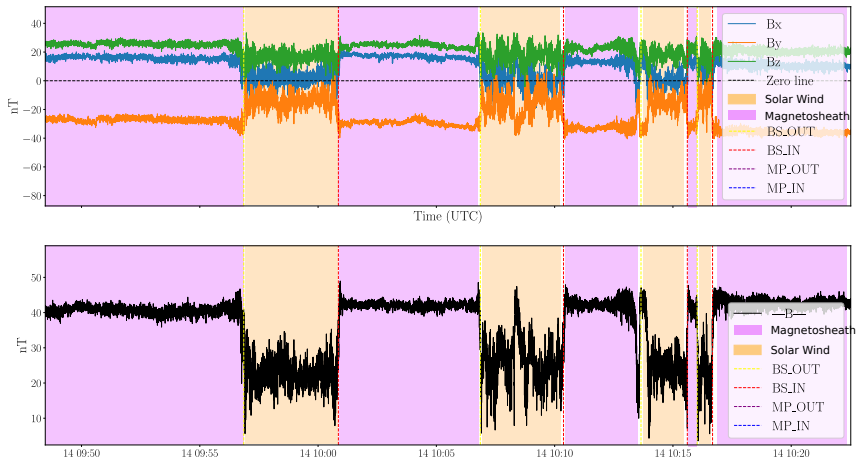
Regions, Crossings, & Encounters on MAG Data

MESSENGER MAG data, taken from 2012-10-14-2012-10-14



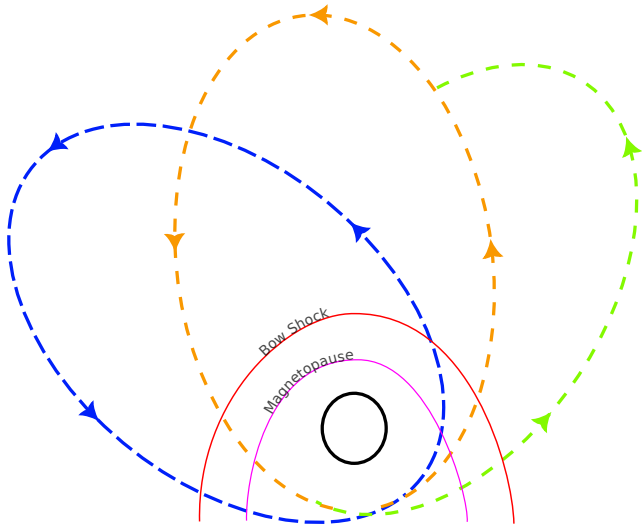
Regions, Crossings, & Encounters on MAG Data

MESSENGER MAG data, taken from 2012-10-14-2012-10-14

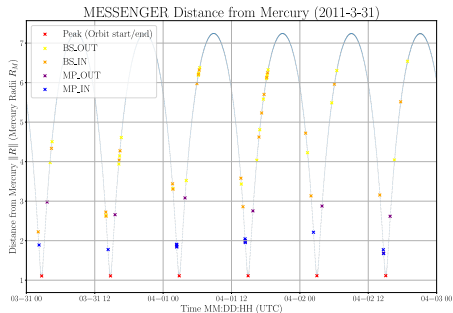
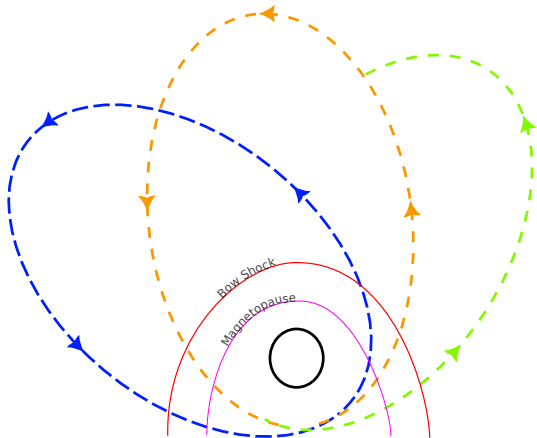


Defining an Orbit

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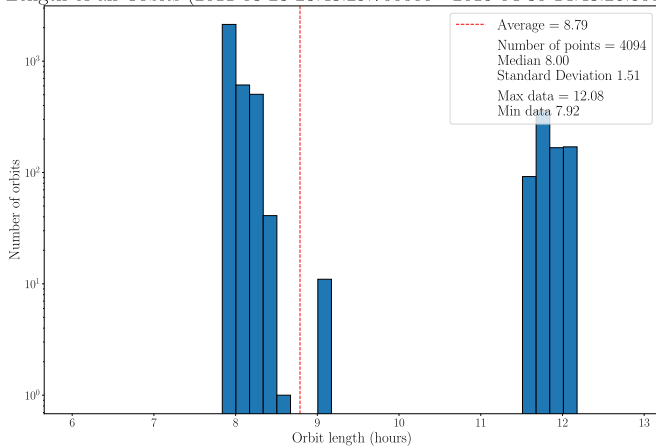
Defining an Orbit



Plot of radial distance from Mercury over a few days. The red crosses mark the periapsides, while the rest of the crosses mark the type of boundary crossing.

Defining an Orbit

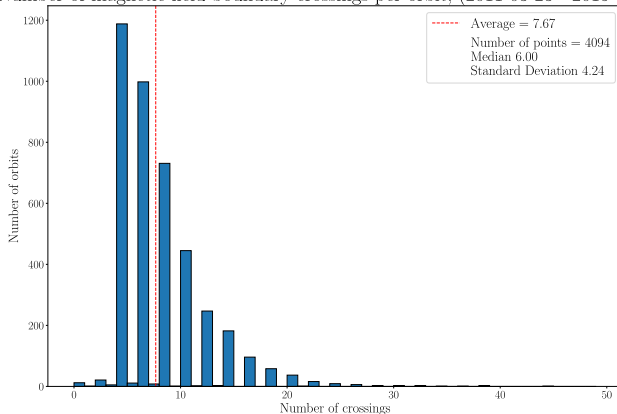
Length of all Orbits (2011-03-23 23:48:29.700000 - 2015-04-30 14:43:26.500000)



Number of Crossings per Orbit

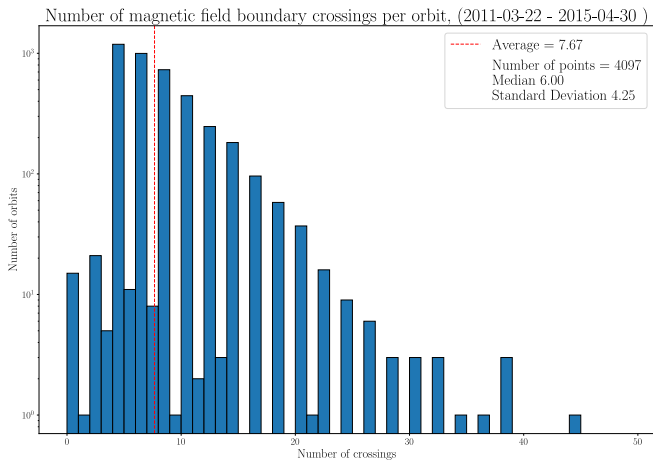
Number of Crossings per Orbit

Number of magnetic field boundary crossings per orbit, (2011-03-23 - 2015-04-30)

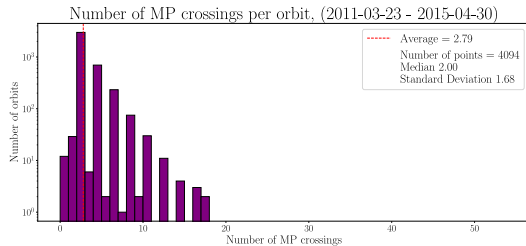
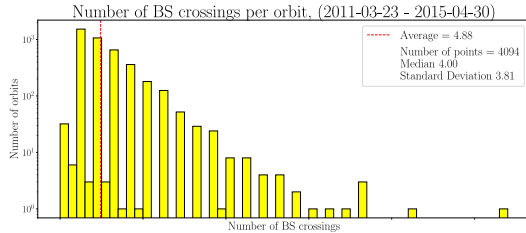


Histogram of number of crossing per orbit using Hollman's 2025 list of crossings, determined with a region classifier machine learner algorithm.

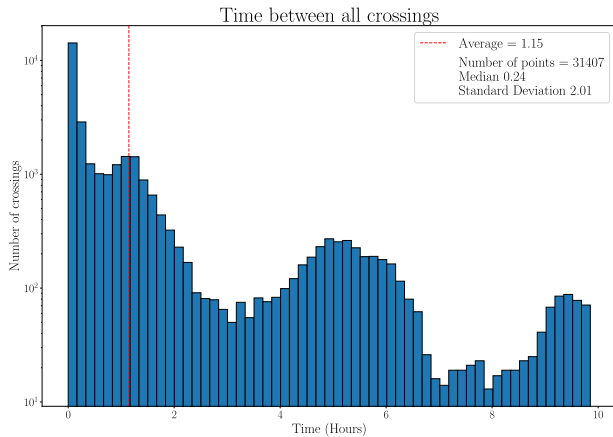
Number of Crossing per Orbit



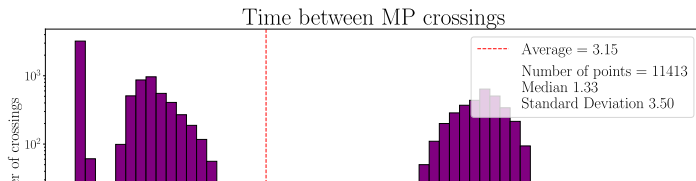
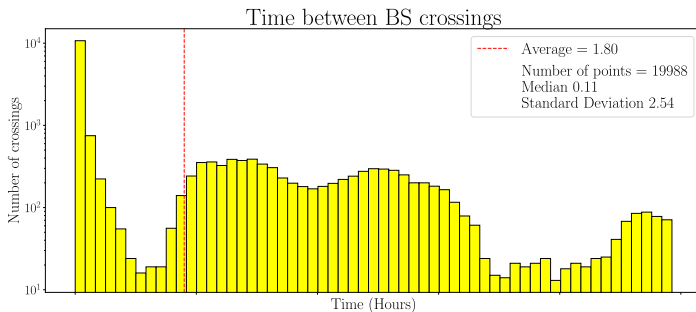
Number of Each Crossing Type per Orbit



Time Between Crossings



Time Between Each Crossing Type



Time Between Encounters

